TECHOFAY

AI-POWERED GYM MANAGEMENT & BUSINESS GROWTH PLATFORM

Scope of Work & Developer Requirements

Prepared for product planning, software development, implementation and SaaS commercialization

Product Positioning

A scalable, multi-tenant SaaS platform combining Gym Management, Memberships, Attendance, Billing, CRM, Fitness Management, Marketing Automation, Analytics and AI.

“Manage your gym. Grow your members. Automate your business.”

1. Project Objective

- Design and develop a powerful, secure and scalable gym-management platform for gyms, fitness centres, studios, personal training centres and multi-branch fitness chains.
- Build the system as an API-first, modular and multi-tenant SaaS product so Techofay can onboard and manage multiple independent gym businesses.
- Provide a strong alternative to established gym-management platforms by combining operational management with CRM, marketing automation, advanced analytics and AI.

2. Core User Roles

- Super Admin / SaaS Administrator
- Gym Owner
- Branch Manager
- Receptionist
- Trainer / Personal Trainer
- Nutritionist
- Sales / Business Development Executive
- Accountant
- Gym Member

3. Super Admin & SaaS Administration

- Gym/organization onboarding, activation, suspension and tenant management
- Subscription plans, trials, billing and usage monitoring
- Platform-wide analytics, revenue dashboard and system configuration
- Feature flags, support/ticket management and audit logs
- Tenant-level settings with strict data isolation

4. Gym Owner Dashboard

- Real-time KPIs: members, active/expired memberships, attendance, revenue, dues, leads and renewals
- Revenue, membership growth, attendance, lead-funnel and retention charts
- Trainer performance, branch comparison and business alerts

5. Member Management

- Online/reception registration, unique Member ID and QR code
- Profile, photograph, emergency contact, documents and fitness information
- Membership, payment, attendance, workout, diet, trainer and progress history
- Member status: Active, Expired, Suspended and configurable custom statuses

6. Membership Management

- Monthly, quarterly, half-yearly, annual, custom, trial, couple, family and corporate plans
- Personal training packages, discounts, coupons and promotional pricing
- Membership start/end dates, freeze, extension, transfer, upgrade/downgrade and renewal
- Automated expiry and renewal workflows

7. Attendance Management

- QR, Member ID, mobile number and manual check-in/check-out
- Optional biometric/RFID integrations
- Attendance history, peak-hour analytics and inactive-member identification

- Configurable automated follow-up for members with low attendance

8. Billing & Payments

- Cash, UPI, card, net banking and payment gateway support
- Invoices, receipts, GST-compatible billing, partial payments, refunds and adjustments
- Outstanding dues, payment history and automated payment reminders
- Revenue, expense and net-revenue dashboards

9. CRM & Lead Management

- Lead capture from website, Google, social media, WhatsApp, walk-ins, referrals and advertising campaigns
- Pipeline: New → Contacted → Interested → Trial Booked → Trial Completed → Converted → Lost
- Lead assignment, follow-ups, notes, call history, WhatsApp/email communication
- Lead-source attribution, conversion tracking and sales-representative performance

10. Trial Class & Booking Management

- Trial registration, slot booking, trainer allocation and attendance
- PT, group-class and trainer booking with capacity, cancellation and waitlist support
- Automated confirmations, reminders and trial-to-member conversion tracking

11. Trainer & Personal Training Management

- Trainer profiles, qualifications, certifications, specializations and availability
- Assigned-member dashboard, sessions, workout/diet plans and member progress
- PT packages, session tracking, remaining sessions, expiry and trainer commission

12. Workout & Exercise Management

- Exercise library with muscle group, difficulty, instructions, images/videos and equipment
- Daily/weekly/monthly workout plans with sets, repetitions, weight and rest time
- Trainer-created personalized programs and member progress tracking

13. Nutrition & Diet Management

- Nutrition profile, calorie targets, meal schedules, food database and macronutrient tracking
- Water intake, diet notes and progress history
- AI-assisted preliminary diet recommendations with professional review controls

14. Fitness Assessment & Progress

- Weight, height, BMI, body fat, muscle mass and configurable body measurements
- Measurement history, graphs, progress photos, goals and trainer observations
- Before/after comparison and progress dashboard

15. Communication & Notifications

- WhatsApp Business API, SMS, email, in-app and push notifications
- Expiry, renewal, payment, birthday, trial, appointment, inactivity and promotional messages
- Template management and communication history

16. Marketing Automation

- New-member, renewal, win-back, festival, referral, trial and birthday campaigns
- Audience segmentation by membership, attendance, age, location, spending, inactivity and fitness goals
- Campaign performance and conversion reporting

17. AI-Powered Features

- AI member-retention / churn risk insights
- AI lead scoring: Hot / Warm / Cold
- AI follow-up recommendations and message generation
- AI business insights and natural-language reporting
- AI-assisted workout recommendations
- AI chatbot for memberships, timings, pricing, trainers, trials and common support queries

18. Expense & Inventory Management

- Expense categories, vendors, recurring expenses, salary, utilities, equipment and marketing expenses
- Inventory for supplements, merchandise, apparel, accessories and equipment
- Products, stock in/out, purchase, sales, suppliers and low-stock alerts

19. Staff & Role Management

- Staff profiles, attendance, permissions, performance, commissions and salary records
- Role-based access control with configurable module permissions

20. Multi-Branch Management

- Centralized management of multiple gym branches
- Branch-wise members, revenue, attendance, expenses, staff and performance
- Cross-branch comparison and consolidated reporting

21. Member Portal / PWA

- Responsive web/PWA access on desktop, tablet and mobile
- Member profile, membership, QR code, attendance, workout, diet, trainer, payments, invoices, notifications and progress

22. Website & Lead Integration

- Public website pages for plans, trainers, facilities, gallery, testimonials and contact
- Online trial booking and membership enquiry forms
- Website leads automatically pushed into CRM
- Google Maps, WhatsApp and payment integration

23. Reports & Analytics

- Member, membership, revenue, payment, attendance, trainer, PT, lead, conversion, renewal, churn and expense reports
- Export to PDF, Excel and CSV
- Role-based dashboards and configurable date/branch filters

24. Security & Compliance Requirements

- HTTPS/SSL, secure authentication, password hashing and token/session security
- Role-based access control, API authentication and rate limiting
- Secure file upload, audit logs, login history, backups and disaster-recovery procedures
- Tenant data isolation and least-privilege access
- Privacy controls for personal and fitness information

25. Recommended Technology Stack

Layer	Recommended Technology	Purpose
Frontend	React.js / Next.js + TypeScript + Tailwind CSS	Fast, responsive, maintainable web application and PWA
Backend	Node.js + NestJS + TypeScript	Scalable modular APIs and business logic
Database	PostgreSQL	Transactional core data and multi-tenant architecture
Caching / Queues	Redis	Caching, sessions, rate limits and background jobs
Search	OpenSearch / Elasticsearch (optional)	Advanced search and analytics
AI Layer	LLM API + AI orchestration + optional vector DB	AI assistant, insights, lead scoring and knowledge retrieval
Cloud	AWS / Azure / Google Cloud	Production hosting, storage, networking and monitoring
DevOps	Docker + CI/CD; Kubernetes when scale requires	Repeatable deployment and horizontal scaling
Storage	Object storage + CDN	Member documents, images, videos and backups

26. Recommended Integrations

- Payment gateway such as Razorpay or equivalent
- WhatsApp Business API provider
- SMS gateway and transactional email provider
- Google Maps and authentication providers where required
- Accounting/ERP APIs where required
- Biometric, RFID and QR attendance hardware
- Analytics and monitoring platforms

27. SaaS Architecture & Commercial Model

- Multi-tenant architecture with isolated gym organizations and configurable tenant settings
- Central Super Admin console for onboarding, subscription plans, feature controls and usage monitoring
- Suggested SaaS tiers: Starter, Professional and Enterprise
- Architecture should support white-label deployments and future mobile applications

28. Development Phases

Phase	Scope
1. Discovery & Architecture	Requirements, workflows, database model, UI/UX, API and technical architecture
2. Core Management	Admin, members, memberships, attendance, billing and staff
3. CRM & Fitness	Leads, sales pipeline, trainers, PT, workout, nutrition and assessments
4. Automation	WhatsApp, SMS, email, notifications and marketing automation
5. AI & Analytics	AI assistant, lead scoring, retention insights and advanced dashboards
6. SaaS & Multi-Branch	Tenant management, subscription plans, branches, Super Admin and usage analytics
7. QA & Deployment	Functional, API, security and performance testing, UAT, deployment and documentation

29. Competitive Positioning

The product should compete with established gym-management solutions such as OKFIT and comparable platforms while differentiating through business-growth capabilities.

Capability	Techofay Platform
Gym & Member Management	✓
Membership & Attendance	✓
Billing & Payments	✓
CRM & Lead Pipeline	✓
WhatsApp & Marketing Automation	✓
Trainer / PT / Workout / Nutrition	✓
Multi-Branch & SaaS	✓
Advanced Business Analytics	✓
AI Lead Scoring & Retention Insights	✓
AI Business Assistant / Chatbot	✓
API-First & White-Label Potential	✓

30. Developer Acceptance Principles

- Modular codebase with documented APIs and database schema
- Responsive UI across desktop, tablet and mobile
- Role-based permissions enforced at API and UI levels
- Automated testing for critical workflows
- Logging, monitoring and error tracking enabled in production
- Backup and recovery procedures documented
- No cross-tenant data leakage

- Third-party integrations isolated behind service interfaces to allow provider changes
- AI features must include configurable safeguards and human-review controls where recommendations affect fitness or nutrition decisions

TECHOFAY GYMO S

AI-powered Gym Management & Business Growth Platform

Built for operational efficiency today and scalable SaaS growth tomorrow.